

Module 05: Action in Embodied

Learning Objectives

1. Define **Action** within the context of Embodied.
2. Analyze how Action interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Action.

Introduction

This module explores **Action**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Action is a critical component of the 8-part Active Inference spine, bridging the gap between Cognition and Learning.

Key Concepts

1. Action as a Markov Blanket Boundary

How does Action define the boundary between the agent and the environment?

2. Generative Models of Action

What parameters involved in Action must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Action drive the perception-action loop?

Applications

In Embodied, we see Action manifest in: * **Specific Example 1:** When a capoeira practitioner executes a ginga -- the fundamental rocking movement between stances -- her body flows

through a continuous loop of weight shifts, arm swings, and torso rotations that are simultaneously defensive, offensive, and expressive; this skilled action is active inference at full embodiment, where the generative model predicts the proprioceptive trajectory of the entire movement sequence and the body follows the predicted path, generating smooth coordinated motion that minimizes free energy across the musculoskeletal system. * **Specific Example 2:** A surfer dropping into a wave makes split-second adjustments of ankle pressure, knee bend, hip rotation, and arm position to maintain balance on a surface that is itself moving and changing shape; each micro-action generates new proprioceptive and vestibular feedback that the generative model uses to predict the wave's next moment, creating a continuous action-perception cycle where the body actively sculpts its relationship with the moving environment to minimize the expected surprise of falling.

Conclusion

Understanding Action allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.